PROJECT FILE

Web Programming using

Python and JavaScript

Submitted by:

Name: Aarushi singh

Course: B.Tech CSE Core

Roll no:2401010002

Submitted to:

Name: Dr. Mansi Kajal

Experiment-1

Aim: Personal Portfolio Website using HTML, CSS Styling and JavaScript.

Tasks:

- Project Setup & Introduction – Create Project folder and Basic linked files.
- HTML Structure – Build portfolio section using semantic HTML.
- Styling with CSS – Apply layout, colors, spacing, and neat design.
- Projects & Skills Display – Show skills and projects clearly on the page.
- Contact Form – Create a form with user submission feedback.

Tech Stack Used:

- HTML, CSS, and JavaScript

CODES:

HTML Code:

portfolio > port.html > html > body

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Varsha Rani | Portfolio</title>

  <!-- Google Font -->
  <link href="https://fonts.googleapis.com/css2?family=Poppins:wght@400;500;600;700&display=swap" rel="stylesheet">

  <link rel="stylesheet" href="style.css" />
</head>
<body>

  <!-- NAVBAR -->
  <header>
    <nav class="navbar">
      <h2 class="logo">Varsha Rani</h2>

      <ul class="nav-links">
        <li><a href="#home" class="nav-item active">Home</a></li>
        <li><a href="#about" class="nav-item">About</a></li>
        <li><a href="#skills" class="nav-item">Skills</a></li>
        <li><a href="#contact" class="nav-item">Contact</a></li>
      </ul>
    </nav>
  </header>

  <!-- HERO -->
  <section class="hero" id="home">
    <div class="hero-content">
      <h1>Hi, I'm <span>Varsha Rani</span></h1>
      <h2>B.Tech CSE Student | Web Developer</h2>
      <p>I am a passionate web developer...</p>
      <a href="#contact" class="btn">Contact Me</a>
    </div>
  </section>
```

```html
<html lang="en">
<body>

  <!-- ABOUT -->
  <section class="about" id="about">
    <h2 class="title">About Me</h2>
    <p class="text">
      I am <b>Varsha Rani</b>, a B.Tech CSE student and a passionate Web Developer.
      I love building responsive and user-friendly websites. I am learning and building
      projects using modern web technologies.
    </p>
  </section>

  <!-- SKILLS -->
  <section class="skills" id="skills">
    <h2 class="title">My Skills</h2>

    <div class="skills-box">
      <div class="skill">HTML</div>
      <div class="skill">CSS</div>
      <div class="skill">JavaScript</div>
      <div class="skill">C++</div>
      <div class="skill">Firebase</div>
    </div>
  </section>

  <!-- CONTACT -->
  <section class="contact" id="contact">
    <h2 class="title">Contact Me</h2>

    <form class="contact-form" id="contactForm">
      <input type="text" id="name" placeholder="Your Name" required />
      <input type="email" id="email" placeholder="Your Email" required />
      <textarea id="message" rows="5" placeholder="Your Message" required></textarea>

      <button type="submit" class="btn">Send Message</button>
      <p id="formMsg"></p>
```

```html
      <button type="submit" class="btn">Send Message</button>
      <p id="formMsg"></p>
    </form>
  </section>

  <!-- FOOTER -->
  <footer>
    <p>© 2026 Varsha Rani | All Rights Reserved</p>
  </footer>

  <!-- JavaScript -->
  <script src="script.js"></script>

</body>
</html>
```

CSS Code:

```css
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
  font-family: "Poppins", Arial, sans-serif;
}

html {
  scroll-behavior: smooth;
}

body {
  background: linear-gradient(135deg, #eaf6ff, #ffffff);
  color: #0b1220;
  overflow-x: hidden;
}

/* ===== NAVBAR ===== */
header {
  position: sticky;
  top: 0;
  z-index: 1000;
  padding: 15px 8%;
  background: rgba(255, 255, 255, 0.75);
  backdrop-filter: blur(12px);
  border-bottom: 1px solid rgba(0, 0, 0, 0.06);
}

.navbar {
  display: flex;
  justify-content: space-between;
  align-items: center;
}

.logo {
  color: #009dff;
```

```
.logo {
  color: #009dff;
  font-size: 24px;
  font-weight: 700;
  letter-spacing: 1px;
  text-transform: uppercase;
}

/* Nav Links */
.nav-links {
  list-style: none;
  display: flex;
  gap: 25px;
}

.nav-links a {
  text-decoration: none;
  color: #0b1220;
  font-size: 16px;
  font-weight: 500;
  position: relative;
  transition: 0.3s;
}

.nav-links a::after {
  content: "";
  position: absolute;
  left: 0;
  bottom: -6px;
  width: 0%;
  height: 2px;
  background: #009dff;
  transition: 0.3s;
}
```

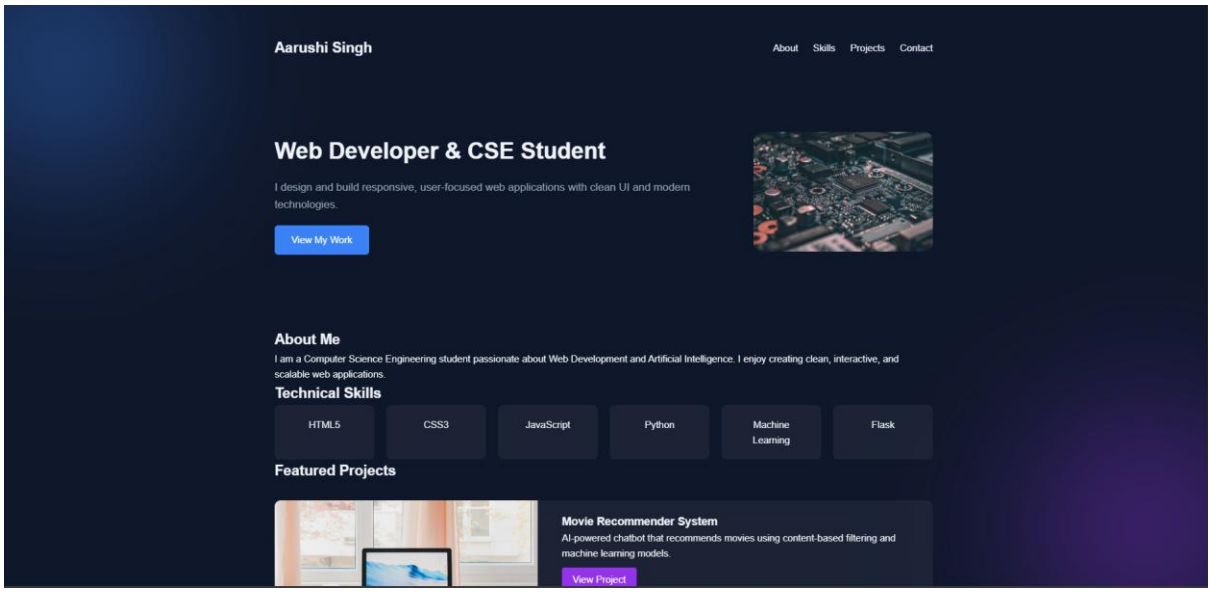
```
  color: #009dff;
}

.nav-links a:hover::after {
  width: 100%;
}

/* ===== HERO SECTION ===== */
.hero {
  min-height: 92vh;
  display: flex;
  justify-content: center;
  align-items: center;
  text-align: center;
  padding: 0 8%;
  position: relative;
}

/* Soft glow background circles */
.hero::before,
.hero::after {
  content: "";
  position: absolute;
  width: 300px;
  height: 300px;
  border-radius: 50%;
  filter: blur(90px);
  opacity: 0.45;
  z-index: -1;
}

.hero::before {
  background: #8fd3ff;
  top: 10%;
  left: -5%;
}
```

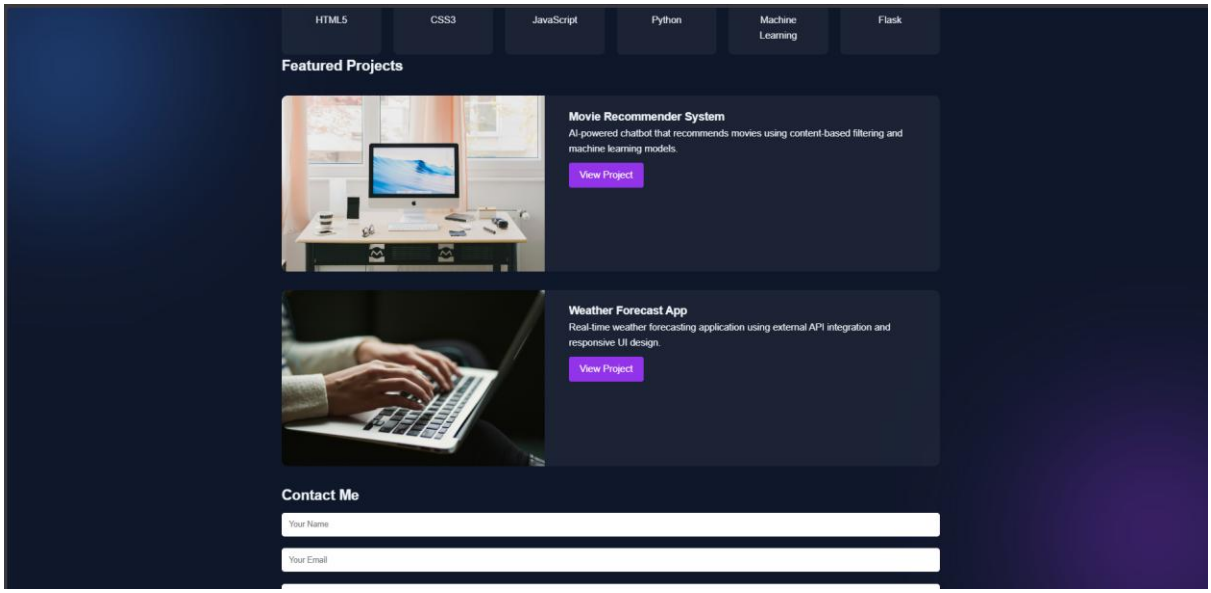
```
.hero::before {
  background: #8fd3ff;
  top: 10%;
  left: -5%;
}

.hero::after {
  background: #ccecff;
  bottom: 10%;
  right: -5%;
}

/* Hero Card */
.hero-content {
  max-width: 750px;
  padding: 50px 30px;
  border-radius: 18px;
  background: rgba(255, 255, 255, 0.85);
  border: 1px solid rgba(0, 0, 0, 0.05);
  box-shadow: 0 12px 35px rgba(0, 0, 0, 0.08);
  animation: fadeInUp 1s ease;
}

.hero-content h1 {
  font-size: 52px;
  margin-bottom: 12px;
  font-weight: 700;
}

.hero-content span {
  color: #009dff;
  text-shadow: 0 0 15px rgba(0, 157, 255, 0.25);
}

.hero-content h2 {
  font-size: 20px;
  color: #445;
```

JavaScript Code:

```
document.getElementById("contactForm").addEventListener("submit", function(e) {
    e.preventDefault()
    document.getElementById("formStatus").innerText = "Thank you! Your message has been submitted."
    this.reset()
})
```

Output:





Weather Forecast App

Real-time weather forecasting application using external API integration and responsive UI design.

View Project

Contact Me

Your Name

Your Email

Your Message

Send Message

© 2026 Aarushi Singh | All Rights Reserved

Experiment-2

Aim: Build a Company Landing Page using HTML and CSS.

Tasks:

- Project Setup & Introduction – Create a project folder with required files index.html and style.css
- HTML Structure – Building HTML structure including Hero Section, About section, Services, Testimonials, Contact Information.
- Styling with CSS – Apply CSS for Page layout, colors/fonts, alignment, margin/padding.
- Service Section Design – Adding at least 3 services, including service name, short description
- Testimonials – Adding at least 2 customer testimonials including customer name, and their feedback.
- Contact Form – Adding a contact form with address/email/phone

Codes:

HTML Code:

Company page > company.html > html > body > header.navbar > nav > ul.nav-links > li

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>RK Properties | Real Estate</title>
  <link rel="stylesheet" href="company.css" />
</head>
<body>

  <!-- Navbar -->
  <header class="navbar">
    <div class="logo">RK Properties</div>
    <nav>
      <ul class="nav-links">
        <li><a href="#home">Home</a></li>
        <li><a href="#services">Properties</a></li>
        <li><a href="#about">About</a></li>
        <li><a href="#contact">Contact</a></li>
      </ul>
    </nav>
    <a href="#contact" class="btn">Book Visit</a>
  </header>

  <!-- Hero Section -->
  <section class="hero" id="home">
    <div class="hero-content">
      <h1>Find Your Dream Home With RK Properties</h1>
      <p>
        We help you buy, sell, and rent properties with trust and transparency.
        Explore premium homes, flats, and commercial spaces at the best price.
      </p>
      <div class="hero-buttons">
        <a href="#services" class="btn primary">Explore Properties</a>
        <a href="#about" class="btn secondary">Know More</a>
      </div>
    </div>
```

```html
      </div>
    </div>
    <div class="hero-image">
      <img src="https://images.unsplash.com/photo-1564013799919-ab600027ffc6?auto=format&fit=crop&w=900&q=80" alt="Property" />
    </div>
  </section>

  <!-- Services Section -->
  <section class="services" id="services">
    <h2>Our Properties & Services</h2>
    <p class="subtitle">Best real estate solutions for your needs.</p>

    <div class="service-grid">
      <div class="card">
        <h3>Buy Property</h3>
        <p>Choose from premium flats, villas, and plots at affordable rates.</p>
      </div>
      <div class="card">
        <h3>Rent Property</h3>
        <p>Find rental homes and offices in top locations with full support.</p>
      </div>
      <div class="card">
        <h3>Sell Property</h3>
        <p>Sell your property quickly with trusted deals and best market value.</p>
      </div>
    </div>
  </section>

  <!-- About Section -->
  <section class="about" id="about">
    <div class="about-content">
      <h2>About RK Properties</h2>
```

```html
      <div class="about-content">
        <h2>About RK Properties</h2>
        <p>
          RK Properties is a trusted real estate company helping customers find
          the perfect home and investment opportunities. We provide verified
          properties with complete documentation and support.
        </p>
```

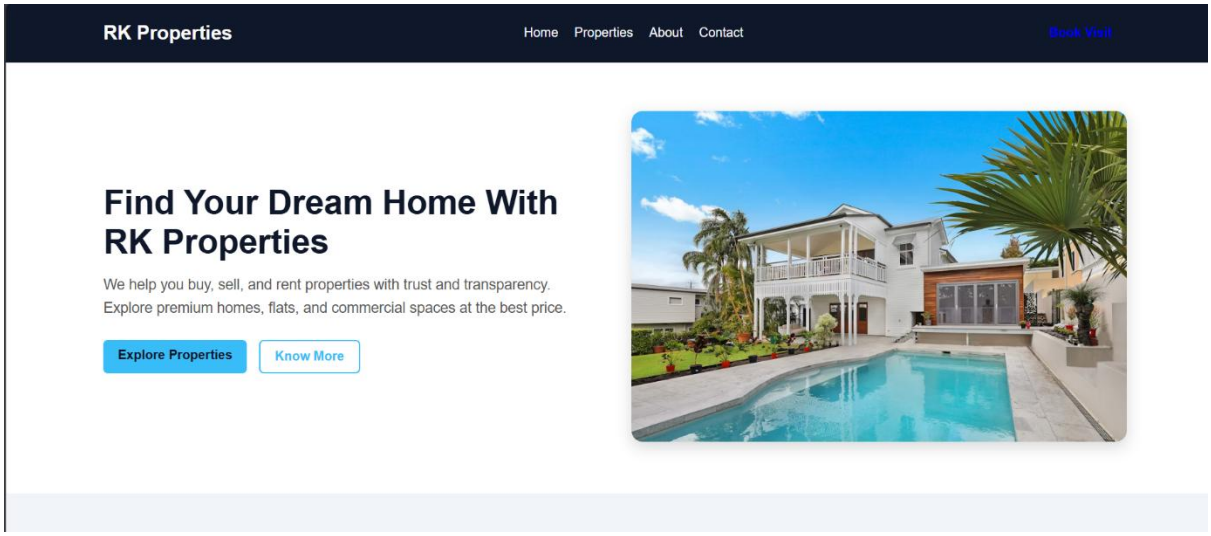
CSS Code:

```css
.logo {
  color: #009dff;
  font-size: 24px;
  font-weight: 700;
  letter-spacing: 1px;
  text-transform: uppercase;
}

/* Nav Links */
.nav-links {
  list-style: none;
  display: flex;
  gap: 25px;
}

.nav-links a {
  text-decoration: none;
  color: #0b1220;
  font-size: 16px;
  font-weight: 500;
  position: relative;
  transition: 0.3s;
}

.nav-links a::after {
  content: "";
  position: absolute;
  left: 0;
  bottom: -6px;
  width: 0%;
  height: 2px;
  background: #009dff;
  transition: 0.3s;
}
```

```css
body {
    font-family: Arial, sans-serif;
    margin: 0;
    padding: 0;
    background-color: #f4f4f4;
}

.hero {
    background-color: #5a8cbe;
    color: white;
    text-align: center;
    padding: 40px;
}

section {
    background-color: white;
    margin: 20px;
    padding: 20px;
    border: 1px solid #ddd;
}

#services {
    display: block;
}

.service-card {
    border: 1px solid #ccc;
    padding: 15px;
    margin-bottom: 15px;
}

#testimonials .testimonial {
    background-color: #eef2f7;
    padding: 15px;
    margin-bottom: 10px;
}

#contact form {
```

```css
#contact form {
    margin-top: 15px;
}

#contact input,
#contact textarea {
    width: 100%;
    margin-bottom: 10px;
    padding: 8px;
}

#contact button {
    padding: 10px 20px;
    background-color: #1f2933;
    color: white;
    border: none;
}

footer {
    text-align: center;
    padding: 15px;
    background-color: #1f2933;
    color: white;
}
```

Output:



Our Properties & Services

Best real estate solutions for your needs.

Buy Property

Choose from premium flats, villas, and plots at affordable rates.

Rent Property

Find rental homes and offices in top locations with full support.

Sell Property

Sell your property quickly with trusted deals and best market value.

About RK Properties

RK Properties is a trusted real estate company helping customers find the perfect home and investment opportunities. We provide verified properties with complete documentation and support.

Experiment-3

Aim: Build a Interactive Quiz Platform using HTML, CSS and JavaScript

Tasks:

- Project Setup and Introduction - Create project folder with required files quiz.html, quiz.css and quiz.js.
- HTML Structure - Building HTML structure including Question text, Multiple Choice options, submit/next button.
- Styling with CSS - Apply CSS for Page layout, colors/fonts, alignment, margin/padding.
- Logic using JavaScript - Building logic and Interactivity of the Platform, capturing answer, checking them, button working, track and update score.
- Displaying Result - Displaying Final score after quiz completion, messages such as excellent, good hide quiz interface after submission. Restart Feature

Codes:

HTML Code:

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Interactive Quiz</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <main class="quiz-container">
        <section id="intro">
            <h1>Interactive Quiz</h1>
            <p>Test your basic web development knowledge. Select the correct answer and move
                through the quiz to see your final score.</p>
            <button id="startBtn">Start Quiz</button>
        </section>

        <section id="quiz" class="hidden">
            <h2 id="question"></h2>
            <form id="options"></form>
            <button id="nextBtn">Next</button>
        </section>

        <section id="result" class="hidden">
            <h2 id="score"></h2>
            <p id="feedback"></p>
            <button id="restartBtn">Restart Quiz</button>
        </section>
    </main>

    <script src="script.js"></script>
</body>
</html>
```


```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Interactive Quiz</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <main class="quiz-container">
        <section id="intro">
            <h1>Interactive Quiz</h1>
            <p>Test your basic web development knowledge. Select the correct answer and move
                through the quiz to see your final score.</p>
            <button id="startBtn">Start Quiz</button>
        </section>

        <section id="quiz" class="hidden">
            <h2 id="question"></h2>
            <form id="options"></form>
            <button id="nextBtn">Next</button>
        </section>

        <section id="result" class="hidden">
            <h2 id="score"></h2>
            <p id="feedback"></p>
            <button id="restartBtn">Restart Quiz</button>
        </section>
    </main>

    <script src="script.js"></script>
</body>
</html>
```

CSS Code:

```
body {
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    background: linear-gradient(135deg, #0f0c29, #302b63, #24243e);
    display: flex;
    justify-content: center;
    align-items: center;
    min-height: 100vh;
    margin: 0;
}

.quiz-container {
    background: linear-gradient(180deg, #0b0f1a, #111827);
    padding: 36px;
    width: 430px;
    border-radius: 14px;
    box-shadow: 0 0 30px rgba(236, 72, 153, 0.35),
                0 0 60px rgba(234, 179, 8, 0.15);
    animation: glowIn 0.7s ease;
}

h1 {
    text-align: center;
    font-size: 28px;
    letter-spacing: 1px;
    background: linear-gradient(90deg, #facc15, #ec4899);
    -webkit-background-clip: text;
    -webkit-text-fill-color: transparent;
    margin-bottom: 12px;
}

h2 {
    text-align: center;
    color: #e5e7eb;
}
```

```
p {
    text-align: center;
    color: #9ca3af;
    margin-bottom: 26px;
}

form {
    margin-top: 20px;
}

label {
    display: flex;
    align-items: center;
    padding: 14px 16px;
    background: linear-gradient(90deg, #111827, #1f2933);
    margin-bottom: 14px;
    border-radius: 12px;
    cursor: pointer;
    transition: all 0.25s ease;
    border: 1px solid #1f2937;
    color: #e5e7eb;
}

label:hover {
    background: linear-gradient(90deg, #1f2937, #312e81);
    box-shadow: 0 0 18px rgba(250, 204, 21, 0.35);
    transform: translateY(-2px);
}

input {
    margin-right: 14px;
    accent-color: #facc15;
}
```

```css
button {
    width: 100%;
    padding: 14px;
    border: none;
    background: linear-gradient(90deg, #facc15, #ec4899);
    color: #000000;
    font-size: 16px;
    font-weight: 800;
    border-radius: 40px;
    cursor: pointer;
    margin-top: 22px;
    letter-spacing: 1px;
    transition: all 0.3s ease;
}

button:hover {
    box-shadow: 0 0 30px rgba(236, 72, 153, 0.6),
                0 0 60px rgba(250, 204, 21, 0.4);
    transform: scale(1.05);
}

#score {
    font-size: 30px;
    background: linear-gradient(90deg, #22d3ee, #facc15);
    -webkit-background-clip: text;
    -webkit-text-fill-color: transparent;
}

#feedback {
    font-size: 20px;
    font-weight: bold;
    margin-top: 10px;
    background: linear-gradient(90deg, #22d3ee, #ec4899);
    -webkit-background-clip: text;
```

```css
.hidden {
    display: none;
}

@keyframes glowIn {
    from {
        opacity: 0;
        transform: translateY(25px) scale(0.95);
    }
    to {
        opacity: 1;
        transform: translateY(0) scale(1);
    }
}
```

JavaScript:

```javascript
const questions = [
    {
        question: "Which type of data is considered personally identifiable information?",
        options: [
            "Browser type",
            "IP address",
            "Screen resolution",
            "Operating system"
        ],
        answer: 1
    },
    {
        question: "What does end-to-end encryption ensure?",
        options: [
            "Faster internet speeds",
            "Only sender and receiver can read messages",
            "Messages are backed up online",
            "Passwords are stored securely"
        ],
        answer: 1
    },
    {
        question: "Which online activity leaves the largest digital footprint?",
        options: [
            "Reading articles",
            "Using private browsing mode",
            "Posting on social media",
            "Clearing browser cache"
        ],
        answer: 2
    },
    {
        question: "What is the primary purpose of cookies on websites?",
        options: [
```

```
    options: [
        "Block advertisements",
        "Improve hardware performance",
        "Store user preferences and sessions",
        "Prevent hacking attempts"
    ],
    answer: 2
},
{
    question: "Which practice improves personal online privacy the most?",
    options: [
        "Using the same password everywhere",
        "Disabling two-factor authentication",
        "Regularly updating passwords",
        "Sharing location publicly"
    ],
    answer: 2
},
{
    question: "What is metadata?",
    options: [
        "Encrypted data",
        "Data that describes other data",
        "Deleted information",
        "Temporary cache files"
    ],
    answer: 1
},
{
    question: "Why are public Wi-Fi networks considered risky?",
    options: [
        "They are slower than private networks",
        "They reduce device battery life",
```

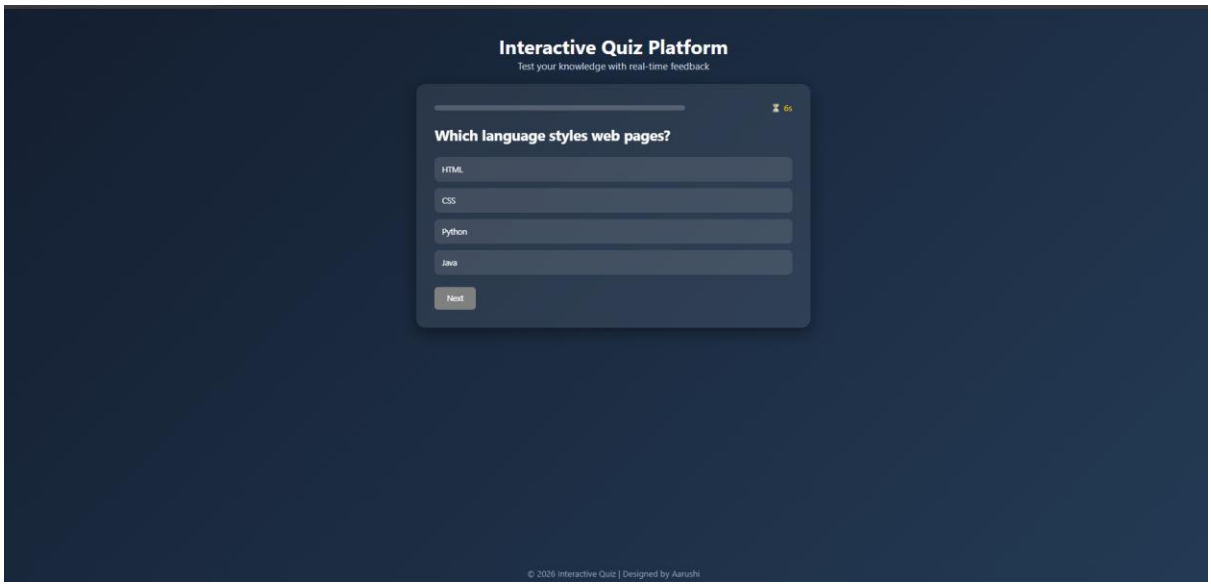
```
        options: [
            "They reduce device battery life",
            "Data can be intercepted by others",
            "They limit website access"
        ],
        answer: 2
    },
    {
        question: "Which tool is commonly used to mask a user's IP address?",
        options: [
            "Firewall",
            "VPN",
            "Antivirus",
            "Cookie blocker"
        ],
        answer: 1
    },
    {
        question: "What does phishing attempt to do?",
        options: [
            "Improve account security",
            "Encrypt personal files",
            "Trick users into revealing sensitive information",
            "Block malicious websites"
        ],
        answer: 2
    },
    {
        question: "Why is two-factor authentication effective?",
        options: [
            "It removes the need for passwords",
            "It adds an extra layer of verification",
            "It speeds up login times",
            "It stores credentials offline"
```

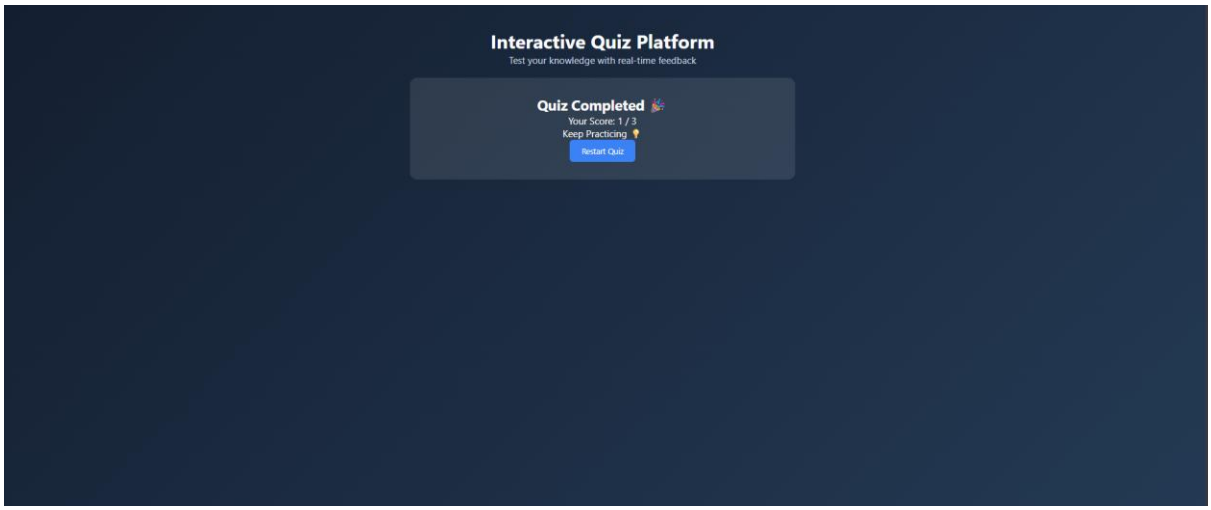
```
        ],
        answer: 1
    }
];

let currentQuestion = 0;
let score = 0;

const intro = document.getElementById("intro");
const quiz = document.getElementById("quiz");
const result = document.getElementById("result");
const questionEl = document.getElementById("question");
const optionsEl = document.getElementById("options");
const nextBtn = document.getElementById("nextBtn");
const scoreEl = document.getElementById("score");
const feedbackEl = document.getElementById("feedback");
const startBtn = document.getElementById("startBtn");
const restartBtn = document.getElementById("restartBtn");

startBtn.addEventListener("click", () => {
    intro.classList.add("hidden");
    quiz.classList.remove("hidden");
    loadQuestion();
});

function loadQuestion() {
    questionEl.textContent = questions[currentQuestion].question;
    optionsEl.innerHTML = "";

    questions[currentQuestion].options.forEach((option, index) => {
        const label = document.createElement("label");
        const input = document.createElement("input");
        input.type = "radio";
        input.name = "option";
```

```
function loadQuestion() {
    questions[currentQuestion].options.forEach((option, index) => {
        label.appendChild(input);
        label.appendChild(document.createTextNode(option));
        optionsEl.appendChild(label);
    });
}

nextBtn.addEventListener("click", () => {
    const selected = document.querySelector("input[name='option']:checked");
    if (!selected) return;

    if (parseInt(selected.value) === questions[currentQuestion].answer) {
        score++;
    }

    currentQuestion++;

    if (currentQuestion < questions.length) {
        loadQuestion();
    } else {
        showResult();
    }
});

function showResult() {
    quiz.classList.add("hidden");
    result.classList.remove("hidden");
    scoreEl.textContent = `Your Score: ${score} / ${questions.length}`;

    if (score >= 8) {
        feedbackEl.textContent = "Excellent awareness!";
    } else if (score >= 5) {
        feedbackEl.textContent = "Good understanding!";
```

```
function showResult() {
    quiz.classList.add("hidden");
    result.classList.remove("hidden");
    scoreEl.textContent = `Your Score: ${score} / ${questions.length}`;

    if (score >= 8) {
        feedbackEl.textContent = "Excellent awareness!";
    } else if (score >= 5) {
        feedbackEl.textContent = "Good understanding!";
    } else {
        feedbackEl.textContent = "Needs improvement!";
    }
}

restartBtn.addEventListener("click", () => {
    currentQuestion = 0;
    score = 0;
    result.classList.add("hidden");
    quiz.classList.remove("hidden");
    loadQuestion();
});
```

Output


Interactive Quiz Platform
Test your knowledge with real-time feedback
6s
Which language styles web pages?
HTML
CSS
Python
Java
Next
© 2026 Interactive Quiz | Designed by Aarushi


Interactive Quiz Platform
Test your knowledge with real-time feedback
Quiz Completed 🎉
Your Score: 1 / 3
Keep Practicing 💪
Restart Quiz